

Math 421: Homework 10

Solutions

Due: Dec 5, 2025

Problem 1: (Ross 18.5 (a)) - Solution given in back of textbook.

Problem 2: (Ross 18.10) Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Prove there exist $x, y \in [0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$.

Proof. Let f be a function continuous on $[0, 2]$ such that $f(0) = f(2)$ and consider the function $g(x) = f(x + 1) - f(x)$ defined on the interval $[0, 1]$. $g(x)$ is a continuous function since it involves composition and subtraction of continuous functions. Note that

$$g(0) = f(1) - f(0) = f(1) - f(2) = -(f(2) - f(1)) = -g(1).$$

Therefore either $g(0) \leq 0 \leq g(1)$ or $g(1) \leq 0 \leq g(0)$. In either case, $g(x)$ satisfies the hypotheses of the Intermediate Value Theorem on the interval $[0, 1]$ so there exists $x \in [0, 1]$ such that $g(x) = 0 = f(x + 1) - f(x)$, let $y = x + 1$, then $x, y \in [0, 2]$, $|y - x| = 1$ and $f(x) = f(y)$. $\square$

Problem 3: (Ross 19.6 (b))

Theorem. $f(x) = \sqrt{x}$ for $x \geq 0$ is uniformly continuous on $[1, \infty)$.

Proof. Let $\varepsilon > 0$ be given. Let $x, y \in [1, \infty)$, then $\sqrt{x} + \sqrt{y} \geq 2$. If $|x - y| < \delta_1 = 2\varepsilon$, then

$$\begin{aligned} |f(x) - f(y)| &= |\sqrt{x} - \sqrt{y}| \\ &= |\sqrt{x} - \sqrt{y}| \cdot \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} \\ &= \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \\ &\leq \frac{|x - y|}{2} \\ &< \frac{\delta_1}{2} = \varepsilon \end{aligned}$$

Thus f is uniformly continuous on $[1, \infty)$. $\square$

Problem 4: (Ross 19.7) - Solution given in back of textbook.

Problem 5 Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that if there exists $M \geq 0$ such that

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Proof. Case 1 ($M = 0$): If $M = 0$, let $\epsilon > 0$ and assume $|x - y|$ is less than any positive number. Then

$$|f(x) - f(y)| \leq 0 < \epsilon.$$

Case 2 ($M > 0$): Let $\epsilon > 0$ and $M > 0$. Assume $|x - y| < \frac{\epsilon}{M}$. Then

$$|f(x) - f(y)| \leq M|x - y| < M \cdot \frac{\epsilon}{M} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, f is uniformly continuous on $\mathbb{R}$. □

Problem 6 Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \rightarrow \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g .

Proof.

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{xg(x) - 0g(0)}{x} = \lim_{x \rightarrow 0} g(x) = g(0).$$

Since g is continuous at $x = 0$. Thus, f is differentiable at $x = 0$ and $f'(0) = g(0)$. □